A Time-Domain Filtering Method Based on Intrapulse Joint Interpulse Coding to Counter Interrupted Sampling Repeater Jamming in SAR

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***Abstract*— The interrupted sampling repeater jamming (ISRJ) can effectively degrade the image quality and affect the subsequent target recognition by creating deceptive multiple false targets on synthetic aperture radar (SAR) images. A time-domain filtering method based on pulse coding to counter ISRJ is proposed in this article. First, this coding method requires the radar to transmit a full pulse signal consisting of multiple subpulse signals several times in the original pulse repetition interval (PRI), and there is a difference in the time distribution of the subpulses transmitted at different moments. Then, use the observation matrix determined by the echo conditions contained in each receiving window to filter the echo in the time domain to obtain the echo corresponding to each subpulse. Finally, the subpulse echo of the jammer sampling section is discarded and the remaining uninterfered subpulse echoes are segmented for pulse compression and subsequent imaging processing to obtain SAR images with a low jamming-to-signal ratio (JSR). Several groups of simulations show that the proposed method is effective against different kinds of ISRJ, and this time-domain filtering method can improve the effect of radar anti-ISRJ and has a high freedom of waveform design.**

***Index Terms*— Interrupted sampling repeater jamming (ISRJ), intrapulse joint interpulse coding, synthetic aperture radar (SAR), time-domain filtering.**

I. Introduction

With the fast growth of digital radio frequency storage technology, coherent interference based on storing and forwarding has brought serious threats to radar. The representative interrupted sampling repeater jamming (ISRJ) can produce multiple fake targets with controllable distribution due to the undersampling characteristic of the jamming signal, which will reduce the quality of synthetic aperture radar (SAR) image and affect the subsequent target detection and recognition of SAR [1], [2], [3]. Moreover, as an intrapulse repeater jamming, the general waveform agility anti-jamming method [4], [5], [6] is invalid for ISRJ. Therefore, it is very necessary to learn the method of anti-ISRJ.

The theory of ISRJ was first proposed in [1], which has attracted much attention from researchers. Literature [7] and [8] extended the theory of ISRJ in range dimension to azimuth dimension and proposed a theory of ISRJ in azimuth to generate multiple false targets on SAR image. The 2-D ISRJ techniques that generate false targets in both the azimuth and range dimensions of SAR images are discussed in [9], [10], and [11]. In [12] and [13], ISRJ is combined with frequency-shifted interference to adjust the location of the false targets. To obtain more realistic interference effects, a method combining ISRJ with scattered wave jamming was proposed in [14].

The current research methods to combat ISRJ are mainly divided into signal processing methods [15], [16], [17], [18], [19], [20], [21], [22] based on time–frequency analysis at the receiver side and waveform design methods at the transmitter side [23], [24], [25], [26], [27], [28], [29], [30], [31], [32]. In [15], [16], and [17], a time–frequency analysis is used to estimate ISRJ parameters, and then, the ISRJ signal is reconstructed for jamming signal cancellation. The suppression of ISRJ by constructing filters in the time–frequency domain was proposed in [18], [19], [20], [21], and [22], but the anti-ISRJ ability of this method will be significantly reduced when the jamming is indistinguishable from the target in terms of range. Wang et al. [23] and Zhou et al. [24], [25] proposed a method for anti-ISRJ with a joint waveform and filter design. Since this method uses an unmatched filtering system, the jamming signal remains residual and the energy of the target is lost. Zhang et al. [31] used the orthogonality between linear frequency modulation (LFM) subpulses to make a narrowband filter bank in the frequency dimension to filter out the signal segments not sampled by the jammer for segmental pulse compression for ISRJ suppression. Then, Zhang et al. [32] improved the LFM signal to a stepped LFM signal to suppress the residual jamming band due to the spectral leakage between subpulses.

Based on the prior knowledge of the jamming parameters, inspired by the frequency-domain filtering method in [31] and [32], we extend the interpulse multipulse coding method

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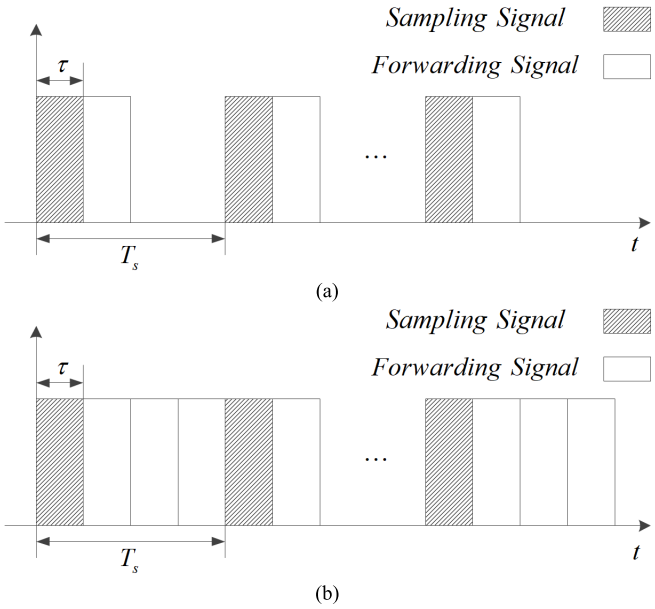


Fig. 1. Interrupted sampling modulated pulse signal. (a) D-ISRJ. (b) R-ISRJ.

previously discussed in [33] with an intrapulse subpulse coding for the purpose of anti-ISRJ. Also, the contribution of the method proposed in this article can be concluded as follows.

1) In order to solve the problem that frequency-domain filtering methods can result in ISRJ not being completely suppressed due to the existence of transition bands between subpulse spectra, a time-domain filtering method is proposed. Complete suppression of the ISRJ can be achieved by obtaining the echo of each subpulse through time-domain filtering and eliminating the echo of the subpulse sampled by the jammer.
2) The method proposed in this article does not affect the imaging focusing quality of the target while performing effective jamming suppression.
3) The coding method has no restriction on the signal parameters of the subpulse, and thus, this method has a high degree of freedom in waveform design.

The rest of this article is arranged as follows. Section II introduces the basic principles of ISRJ. The proposed intrapulse joint interpulse coding signal model and the corresponding time-domain filtering processing method are described in Section III. Then, the influence on the anti-ISRJ ability of the proposed method when the jamming parameter estimation has errors is analyzed quantitatively in Section IV. Section V shows the simulations to demonstrate the availability of the proposed method and its advantages over the frequency-domain filtering method. Finally, some conclusions are noted in Section VI.

II. Principle of ISRJ

ISRJ can produce a realistic false target string jamming effect by sampling and forwarding radar signals at a rate lower than the Nyquist sampling rate.

As shown in Fig. 1, according to different forwarding methods, ISRJ can be divided into direct ISRJ (D-ISRJ) and repeated ISRJ (R-ISRJ). D-ISRJ means that after sampling a small segment of the SAR signal, the sampled signal is forwarded once. Then, it is sampled again after every other sampling cycle and forwarded again, and so on. R-ISRJ means that after sampling each small segment of SAR signal, it is forwarded multiple times until the next sampling period. It is then resampled and retransmitted multiple times and so on. The interrupted sampling modulation signal is periodic square pulses, and it can be written as [1]

$$w(t) = \mathrm{rect}\left(\frac{t}{\tau}\right) \otimes \sum_{m=-\infty}^{+\infty} \delta(t - mT_s) \tag{1}$$

where τ is the sampling pulse duration, T_s is the sampling interval of the jammer, and $\otimes$ is the convolution processing.

By Fourier transformation, the spectrum $W(f)$ of $w(t)$ can be written as [1]

$$\begin{aligned} W(f) &= \sum_{m=-\infty}^{+\infty} \tau f_s \sin c(\pi m f_s)\delta(f - m f_s) \\ &= \sum_{m=-\infty}^{+\infty} a_m \delta(f - m f_s) \end{aligned} \tag{2}$$

where $f_s = 1/T_s$ is the sampling rate of jammer and $a_m = \tau f_s \sin c(\pi m f_s)$.

If the transmitted signal of radar is $x(t)$, then the spectrum $Y_s(f)$ of the ISRJ signal after matched filtering can be expressed as [1]

$$Y_s(f) = \sum_{m=-\infty}^{+\infty} a_m X(f - m f_s) X^*(f). \tag{3}$$

The ISRJ signal after matched filtering is a weighted linear superposition of the matched filter output of the original signal with different frequency shifts mf_s, which can be seen from (3). Then, the time-domain jamming signal $y_{sm}(t)$ can be obtained using the radar signal ambiguity function theory [34], which can be expressed as [1]

$$y_{sm}(t) = \chi(t, -m f_s) \tag{4}$$

where $\chi(\cdot)$ is the radar signal ambiguity function. Substitute the ambiguity function of the LFM waveform into (4), and (4) can be rewritten as [1]

$$\begin{aligned} y_{sm}(t) = \sin c\left[\pi(m f_s + \gamma t)\left(T_p - |t|\right)\right]\left(1 - \frac{|t|}{T_p}\right) \\ \times \exp(j\pi m f_s t) \end{aligned} \tag{5}$$

where T_p is the pulsewidth and γ is the chirp rate.

III. Time-Domain Filtering Anti-ISRJ Method Based on Intrapulse Joint Interpulse Coding Signal

The principle of ISRJ in Section II shows that ISRJ has the characteristics of discontinuous sampling in the time domain. Using this feature and combining it with the idea of cover waveforms, we propose an intrapulse joint interpulse coding waveform and use a time-domain filtering method to efficiently extract the uninterfered signal segments to counter ISRJ.

A. Signal Model of Intrapulse Joint Interpulse Coding

The intrapulse joint interpulse coding method requires the radar to transmit N_p pulses $S_{n_p}(n_p \in [1, N_p])$ within the original pulse repetition interval (PRI), which can be shown in Fig. 2, where each pulse S_{n_p} is composed of N_{sp} subpulses

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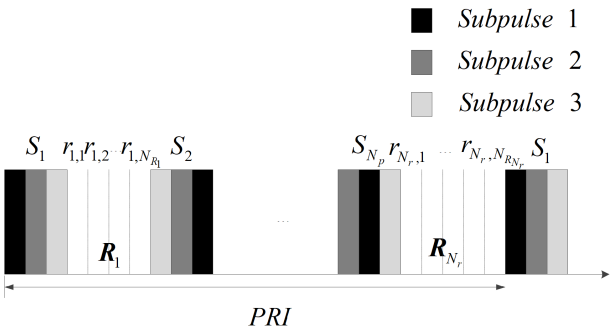


Fig. 2. Intrapulse joint interpulse coding waveform.

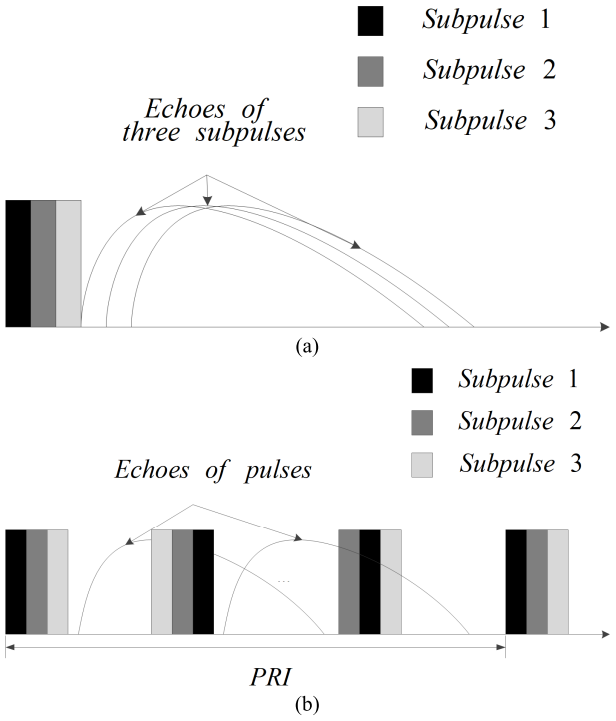


Fig. 3. Echo of coded waveform. (a) Echoes of subpulses. (b) Echoes of pulses.

$s_{n_{sp}}(n_{sp} \in [1, N_{sp}])$ of time width T_{sp} and the time distribution of N_{sp} subpulses $s_{n_{sp}}$ in each pulse signal S_{n_p} can be different. Moreover, the value of T_{sp} is the sampling pulse time width of the ISRJ. $\boldsymbol{R}_{n_r}(n_r \in [1, N_r])$ represents the reception window between each pulse signal and the time width T_{n_r} of each reception window can also be different and needs to satisfy $T_{n_r}/T_{sp} = N_{R_{n_r}}, N_{R_{n_r}} \in N^*$. Then, each reception window $\boldsymbol{R}_{n_r}$ can be expressed as

$$\boldsymbol{R}_{n_r} = \left[r_{n_r,1}, r_{n_r,2}, \ldots, r_{n_r,n_{R_{n_r}}}, \ldots, r_{n_r,N_{R_{n_r}}-1}, r_{n_r,N_{R_{n_r}}}\right], \quad n_{R_{n_r}} \in \left[1, N_{R_{n_r}}\right], \quad n_r \in [1, N_r] \tag{6}$$

where $r_{n_r,n_{R_{n_r}}}$ represents the n_rth subreception window after the $n_{R_{n_r}}$th reception window $\boldsymbol{R}_{n_r}$ has been averaged with T_{sp} as the unit time width.

As shown in Fig. 3(a), since the subpulses are distributed continuously in time, the echoes of the intrapulse subpulses are mixed and missing. Also, it can be seen from Fig. 3(b) that the time interval between pulses becomes smaller due to the increase in the number of transmitted pulses, which causes the echoes of the interpulse subpulses to be overlapped as well. To extract the complete echo of uninterfered subpulses from the mixed and missing echoes for subsequent radar signal processing, a time-domain filtering method is described in detail in the following.

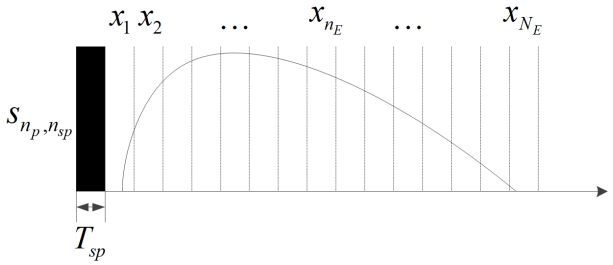


Fig. 4. Echo segmentation.

B. Time-Domain Filtering Method

The time length of echo is limited because of the limited radar beamwidth. For ease of expression, we use $s_{n_p,n_{sp}}$ to represent the subpulse signal $s_{n_{sp}}$ in the n_pth pulse S_{n_p}. As shown in Fig. 4, if the maximum duration of the complete echo $\boldsymbol{X}_{n_p,n_{sp}}$ of subpulse $s_{n_p,n_{sp}}$ from the start of radar reception to the end of reception is T_E, the echo length N_E of the subpulse is defined as

$$N_E = \lceil T_E/T_{sp} \rceil. \tag{7}$$

Then, the echo $\boldsymbol{X}_{n_p,n_{sp}}$ can be expressed as

$$\boldsymbol{X}_{n_p,n_{sp}} = \left[x_{(n_p,n_{sp}),1}, x_{(n_p,n_{sp}),2}, \ldots, x_{(n_p,n_{sp}),n_E}, \ldots, x_{(n_p,n_{sp}),N_E-1}, x_{(n_p,n_{sp}),N_E}\right], \quad n_E \in [1, N_E] \tag{8}$$

where $x_{(n_p,n_{sp}),n_E}$ is the complete echo of the subpulse signal $s_{n_p,n_{sp}}$ in the n_Eth segment of the echo after uniform segmentation by unit time width T_{sp}.

Then, according to the distribution of subpulses in the intrapulse joint interpulse encoded signal, the presence of echoes $\boldsymbol{X}_{n_p,n_{sp}}$ in each reception window $\boldsymbol{R}_{n_r}$ can be characterized as a linear system of equations $\boldsymbol{Q}$ shown in the following equation (9), shown at the bottom of the next page, where $\boldsymbol{R}$ denotes the mixed echoes in the N_r receive windows $\boldsymbol{R}_{n_r}$, $\boldsymbol{X}$ represents the complete echo corresponding to the $N_p \cdot N_{sp}$ subpulse signals, and $\boldsymbol{F}$ is the observation matrix used to represent the true echo condition in each reception window. Each element $\boldsymbol{F}_{n_r,(n_p,n_{sp})}$ in $\boldsymbol{F}$ is used to represent the case of the echo $\boldsymbol{X}_{n_p,n_{sp}}$ present in the n_rth reception window $\boldsymbol{R}_{n_r}$, which can be expanded and written as (10), shown at the bottom of the next page, where $f_{n_{R_{n_r}},n_E}(n_p, n_{sp})$ takes the value of 1 when the n_Eth segment echo $x_{(n_p,n_{sp}),n_E}$ of subpulse signal $s_{n_p,n_{sp}}$ is present in the subreception window $r_{n_r,n_{R_{n_r}}}$. The opposite takes the value of 0.

Take the echo distribution in the reception window $r_{1,2}$ in Fig. 2 as an example to illustrate the process of establishing a system of linear equations. Assuming that the echo length corresponding to the subpulse is 6, then the value of the second line element in matrix $\boldsymbol{F}_{1,(1,2)}$ corresponding to the coded signal in Fig. 2 is [0, 0, 1, 0, 0, 0].

When the radar system parameters satisfy (11) [33]

$$\Delta\varphi = 2\pi V \Delta\theta \Delta t_m/\lambda \le \pi/4 \tag{11}$$

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where $\Delta\varphi$ represents the echo phase difference between the first pulse S_1 and the last pulse S_{N_p} in the coding signal corresponding to the scene center point, V is the speed of radar, λ is the wavelength, Δt_m is the time difference between pulse S_1 and pulse S_{N_p}, and $\Delta\theta$ is the angle of coherent accumulation between the radar transmit S_1 and S_{N_p}. Then, as shown in (12), we can consider that the echoes corresponding to the same subpulse signals in different pulses in the encoded signal are approximately equal

$$\begin{aligned} \boldsymbol{X}_{n_p,n_{\mathrm{sp}}} &= \boldsymbol{X}_{n_{\mathrm{sp}}}, \quad n_p = [1, N_p] \\ &\text{s.t. } \Delta\varphi \le \pi/4. \end{aligned} \tag{12}$$

Then, (9) can be simplified as

$$\begin{aligned} \boldsymbol{Q}_0 : \boldsymbol{F}_0\boldsymbol{X}_0 &= \boldsymbol{R} \\ \boldsymbol{R} &= \left[\boldsymbol{R}_1, \boldsymbol{R}_2, \ldots, \boldsymbol{R}_{n_r}, \ldots, \boldsymbol{R}_{N_r}\right]^T \\ \boldsymbol{X}_0 &= \left[\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_{n_{\mathrm{sp}}}, \ldots, \boldsymbol{X}_{N_{\mathrm{sp}}}\right]^T \\ \boldsymbol{F}_0 &= \begin{bmatrix} \boldsymbol{F}_{1,1} & \boldsymbol{F}_{1,2} & \cdots & \boldsymbol{F}_{1,n_{\mathrm{sp}}} & \cdots & \boldsymbol{F}_{1,N_{\mathrm{sp}}} \\ \boldsymbol{F}_{2,1} & \boldsymbol{F}_{2,2} & \cdots & \boldsymbol{F}_{2,n_{\mathrm{sp}}} & \cdots & \boldsymbol{F}_{2,N_{\mathrm{sp}}} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \boldsymbol{F}_{n_r,1} & \boldsymbol{F}_{n_r,2} & \cdots & \boldsymbol{F}_{n_r,n_{\mathrm{sp}}} & \cdots & \boldsymbol{F}_{n_r,N_{\mathrm{sp}}} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \boldsymbol{F}_{N_r,1} & \boldsymbol{F}_{N_r,2} & \cdots & \boldsymbol{F}_{N_r,n_{\mathrm{sp}}} & \cdots & \boldsymbol{F}_{N_r,N_{\mathrm{sp}}} \end{bmatrix} \end{aligned} \tag{13}$$

where

$$\boldsymbol{F}_{n_r,n_{\mathrm{sp}}} = \sum_{n_p=1}^{N_p} \boldsymbol{F}_{n_r,(n_p,n_{\mathrm{sp}})}. \tag{14}$$

Finally, as shown in (15), the complete echo corresponding to each subpulse can be obtained by solving the system of equations $\boldsymbol{Q}_0$ in (13)

$$\begin{aligned} \boldsymbol{X}_0 &= \boldsymbol{F}_t\boldsymbol{R} \\ \boldsymbol{F}_t &= \left(\boldsymbol{F}_0^T\boldsymbol{F}_0\right)^{-1}\boldsymbol{F}_0^T. \end{aligned} \tag{15}$$

As shown in Fig. 5, the anti-ISRJ method proposed in this article first uses the jamming parameters as prior information for the design of the intrapulse joint interpulse coding signal. Then, use the time-domain filtering matrix $\boldsymbol{F}_t$ to process the overlapped and missing echoes in the receiving window and obtain the complete echo of each subpulse, and then conduct subsection pulse compression processing for the complete echo. Finally, the interfered subpulse echoes are discarded and the uninterfered subpulse echoes are coherently superimposed to perform the subsequent matched filtering in azimuth.

TABLE I
SIMULATION PARAMETERS

Symbol	Parameter	Value
JSR	jamming-to-signal ratio	30 dB
T_s	sampling period	2 us
PRF	PRF	6 KHz
B_{sp}	bandwidth of sub-pulse	10 MHz
T_{sp}	time width of sub-pulse	0.5 us
N_{sp}	number of sub-pulses	8
W_s	swath width	2 km
R_s	slant range of scene center	28 km
H	radar height	14 km
V	radar velocity	380 m/s

$$\begin{aligned} \boldsymbol{Q} : \boldsymbol{F}\boldsymbol{X} &= \boldsymbol{R} \\ \boldsymbol{R} &= \left[\boldsymbol{R}_1, \boldsymbol{R}_2, \ldots, \boldsymbol{R}_{n_r}, \ldots, \boldsymbol{R}_{N_r}\right]^T \\ \boldsymbol{X} &= \left[\boldsymbol{X}_{1,1}, \boldsymbol{X}_{1,2}, \ldots, \boldsymbol{X}_{1,n_{\mathrm{sp}}}, \ldots, \boldsymbol{X}_{1,N_{\mathrm{sp}}}, \boldsymbol{X}_{2,1}, \ldots, \boldsymbol{X}_{n_p,n_{\mathrm{sp}}}, \ldots, \boldsymbol{X}_{N_p,N_{\mathrm{sp}}}\right]^T \\ \boldsymbol{F} &= \begin{bmatrix} \boldsymbol{F}_{1,(1,1)} & \boldsymbol{F}_{1,(1,2)} & \cdots & \boldsymbol{F}_{1,(1,n_{\mathrm{sp}})} & \cdots & \boldsymbol{F}_{1,(1,N_{\mathrm{sp}})} & \boldsymbol{F}_{1,(2,1)} & \cdots & \boldsymbol{F}_{1,(n_p,n_{\mathrm{sp}})} & \cdots & \boldsymbol{F}_{1,(N_p,N_{\mathrm{sp}})} \\ \boldsymbol{F}_{2,(1,1)} & \boldsymbol{F}_{2,(1,2)} & \cdots & \boldsymbol{F}_{2,(1,n_{\mathrm{sp}})} & \cdots & \boldsymbol{F}_{2,(1,N_{\mathrm{sp}})} & \boldsymbol{F}_{2,(2,1)} & \cdots & \boldsymbol{F}_{2,(n_p,n_{\mathrm{sp}})} & \cdots & \boldsymbol{F}_{2,(N_p,N_{\mathrm{sp}})} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \boldsymbol{F}_{n_r,(1,1)} & \boldsymbol{F}_{n_r,(1,2)} & \cdots & \boldsymbol{F}_{n_r,(1,n_{\mathrm{sp}})} & \cdots & \boldsymbol{F}_{n_r,(1,N_{\mathrm{sp}})} & \boldsymbol{F}_{n_r,(2,1)} & \cdots & \boldsymbol{F}_{n_r,(n_p,n_{\mathrm{sp}})} & \cdots & \boldsymbol{F}_{n_r,(N_p,N_{\mathrm{sp}})} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \boldsymbol{F}_{N_r,(1,1)} & \boldsymbol{F}_{N_r,(1,2)} & \cdots & \boldsymbol{F}_{N_r,(1,n_{\mathrm{sp}})} & \cdots & \boldsymbol{F}_{N_r,(1,N_{\mathrm{sp}})} & \boldsymbol{F}_{N_r,(2,1)} & \cdots & \boldsymbol{F}_{N_r,(n_p,n_{\mathrm{sp}})} & \cdots & \boldsymbol{F}_{N_r,(N_p,N_{\mathrm{sp}})} \end{bmatrix} \end{aligned} \tag{9}$$

$$\boldsymbol{F}_{n_r,(n_p,n_{\mathrm{sp}})} = \begin{bmatrix} f_{1,1}(n_p,n_{\mathrm{sp}}) & f_{1,2}(n_p,n_{\mathrm{sp}}) & \cdots & f_{1,n_E}(n_p,n_{\mathrm{sp}}) & \cdots & f_{1,N_E}(n_p,n_{\mathrm{sp}}) \\ f_{2,1}(n_p,n_{\mathrm{sp}}) & f_{2,2}(n_p,n_{\mathrm{sp}}) & \cdots & f_{2,n_E}(n_p,n_{\mathrm{sp}}) & \cdots & f_{2,N_E}(n_p,n_{\mathrm{sp}}) \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ f_{n_{R_{nr}},1}(n_p,n_{\mathrm{sp}}) & f_{n_{R_{nr}},2}(n_p,n_{\mathrm{sp}}) & \cdots & f_{n_{R_{nr}},n_E}(n_p,n_{\mathrm{sp}}) & \cdots & f_{n_{R_{nr}},N_E}(n_p,n_{\mathrm{sp}}) \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ f_{N_{R_{nr}},1}(n_p,n_{\mathrm{sp}}) & f_{N_{R_{nr}},2}(n_p,n_{\mathrm{sp}}) & \cdots & f_{N_{R_{nr}},n_E}(n_p,n_{\mathrm{sp}}) & \cdots & f_{N_{R_{nr}},N_E}(n_p,n_{\mathrm{sp}}) \end{bmatrix}$$

$$f_{n_{R_{nr}},n_E}(n_p,n_{\mathrm{sp}}) = \{0, 1\} \tag{10}$$

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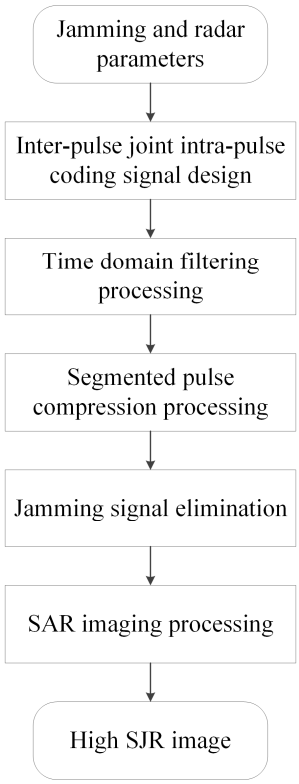


Fig. 5. Algorithm flowchart of time-domain filtering anti-ISRJ.

IV. Analysis of the Influence of Estimation Error of Jamming Parameters on Anti-ISRJ Effectiveness

Since the method for anti-ISRJ of intrapulse joint interpulse coding requires jamming parameters as a priori information, it is essential to analyze the effect of the estimation error of jamming parameters on the anti-ISRJ effectiveness.

As shown in Fig. 6, whether there is an error in the jamming sampling period or the jamming sampling pulsewidth, eventually some original undisturbed subpulses are also sampled and some subpulses are only partially sampled. However, the time-domain filtering algorithm in Section III performs the construction of the system of equations in terms of the echo distribution of a complete subpulse with time width T_{sp}. In this way, (13) is not applicable to represent the jamming distribution in the presence of jamming parameter estimation errors. In the following, the correct linear equation system will be constructed to represent the true distribution of ISRJ echo.

Assuming that there are N_j time widths τ_j, $j \in [1, N_j]$ for each subpulse of radar to be sampled in the case of incorrect estimation of the jamming parameters, a new unit time width T_{unit} different from T_{sp} is defined as

$$T_{\text{unit}} = \max\left\{T | \tau_j / T = M_1, T_{\text{sp}}/T = M_2\right\}, \quad j \in \left[1, N_j\right], \quad M_1, M_2 \in N^*. \tag{16}$$

Then, each subpulse $s_{n_{\text{sp}}}$ can be divided equally into M_2 signals by T_{unit}, where each segment of the signal is represented by $s_{n_{\text{sp}},m}, m \in [1, M_2]$. Also, the length of the complete echo corresponding to each signal $s_{n_{\text{sp}},m}$ can be calculated as

$$N'_E = M_2 \cdot N_E. \tag{17}$$

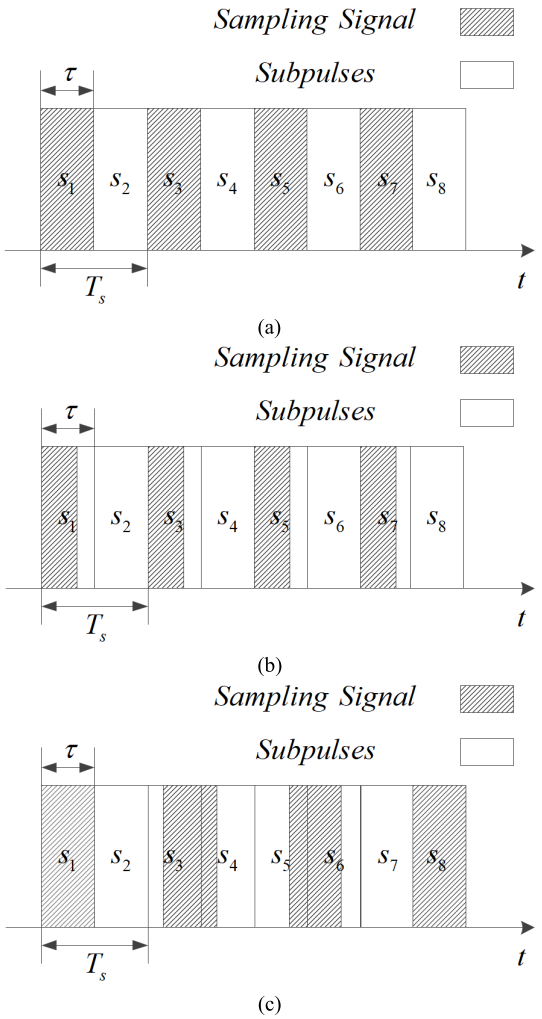


Fig. 6. ISRJ signals under different jamming parameters. (a) Correct jamming parameters. (b) Error in sampling width. (c) Error in sampling period.

Then, the complete jamming echo corresponding to signal $s_{n_{\text{sp}},m}$ can be expressed as

$$\boldsymbol{X}'_{n_{\text{sp}},m} = \left[x'_{(n_{\text{sp}},m),1}, x'_{(n_{\text{sp}},m),2}, \ldots, x'_{(n_{\text{sp}},m),n'_E}, \ldots, x'_{(n_{\text{sp}},m),N'_E}\right], \quad n'_E \in \left[1, N'_E\right] \tag{18}$$

where $x'_{(n_{\text{sp}},m),n'_E}$ is the n'_Eth echo of the complete echo of subpulse signal $s_{n_{\text{sp}},m}$ after uniform segmentation by unit time width T_{unit}.

Similarly, each receive window can be equally divided into $M_2 \cdot N_{R_{n_r}}$ segments with a new unit time width and re-represented as

$$\boldsymbol{R}'_{n_r} = \left[r'_{n_r,1}, r'_{n_r,2}, \ldots, r'_{n_r,n'_{R_{n_r}}}, \ldots, r'_{n_r,N'_{R_{n_r}}}\right], \quad n'_{R_{n_r}} \in \left[1, N'_{R_{n_r}}\right], \quad N'_{R_{n_r}} = M_2 \cdot N_{R_{n_r}}, \ n_r \in [1, N_r] \tag{19}$$

where $r'_{n_r,n'_{R_{n_r}}}$ denotes the $n'_{R_{n_r}}$th subreception window after the n_rth reception window $\boldsymbol{R}_{n_r}$ has been equally divided by T_{unit} as unit time width.

Then, the system of linear equations used to correctly represent the distribution of ISRJ echoes within the reception

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window can be written as (20), shown at the bottom of the page, where each element $\boldsymbol{F}'_{n_r,(n_{\mathrm{sp}},m)}$ in the observation matrix $\boldsymbol{F}_j$ can be expanded and written as (21), shown at the bottom of the page.

Similarly, as $f_{n_{R_{n_r}},n_E}(n_p, n_{\mathrm{sp}})$ in (10), $f'_{n'_{R_{n_r}},n'_E}(n_{\mathrm{sp}}, m)$ takes the value of 1 when the n'_Eth jamming echo $x'_{(n_{\mathrm{sp}},m),n'_E}$ of signal $s_{n_{\mathrm{sp}},m}$ is present in the subreception window $r'_{n_r,n'_{R_{n_r}}}$. The opposite takes the value of 0.

Since (15) is a time-domain filtering process for the echo in the subreception window with time width T_{sp}, while the time width of subreception window in (20) is T_{unit}. Therefore, under the condition that there is an error in the jamming parameter estimation, to obtain the jamming signal after the time-domain filtering shown in (15), the linear system of equations in (20) needs to be deformed to

$$\boldsymbol{Q}'_j : \boldsymbol{F}'_j \boldsymbol{X}'_j = \boldsymbol{R}'_j \tag{22}$$

where

$$\boldsymbol{R}'_j = \begin{bmatrix} \boldsymbol{R}''_{1,1} & \boldsymbol{R}''_{1,2} & \cdots & \boldsymbol{R}''_{1,m} & \cdots & \boldsymbol{R}''_{1,M_2} \\ \boldsymbol{R}''_{2,1} & \boldsymbol{R}''_{2,2} & \cdots & \boldsymbol{R}''_{2,m} & \cdots & \boldsymbol{R}''_{2,M_2} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \boldsymbol{R}''_{n_r,1} & \boldsymbol{R}''_{n_r,2} & \cdots & \boldsymbol{R}''_{n_r,m} & \cdots & \boldsymbol{R}''_{n_r,M_2} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \boldsymbol{R}''_{N_r,1} & \boldsymbol{R}''_{N_r,2} & \cdots & \boldsymbol{R}''_{N_r,m} & \cdots & \boldsymbol{R}''_{N_r,M_2} \end{bmatrix} \tag{23}$$

$$\boldsymbol{X}'_j = \begin{bmatrix} \boldsymbol{X}_j & \cdots & \boldsymbol{0} & \cdots & \boldsymbol{0} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \boldsymbol{0} & \cdots & \boldsymbol{X}_j & \cdots & \boldsymbol{0} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \boldsymbol{0} & \cdots & \boldsymbol{0} & \cdots & \boldsymbol{X}_j \end{bmatrix} \tag{24}$$

$$\boldsymbol{F}'_j = \begin{bmatrix} \boldsymbol{F}''_{1,1} & \boldsymbol{F}''_{1,2} & \cdots & \boldsymbol{F}''_{1,m} & \cdots & \boldsymbol{F}''_{1,M_2} \\ \boldsymbol{F}''_{2,1} & \boldsymbol{F}''_{2,2} & \cdots & \boldsymbol{F}''_{2,m} & \cdots & \boldsymbol{F}''_{2,M_2} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \boldsymbol{F}''_{n_r,1} & \boldsymbol{F}''_{n_r,2} & \cdots & \boldsymbol{F}''_{n_r,m} & \cdots & \boldsymbol{F}''_{n_r,M_2} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \boldsymbol{F}''_{N_r,1} & \boldsymbol{F}''_{N_r,2} & \cdots & \boldsymbol{F}''_{N_r,m} & \cdots & \boldsymbol{F}''_{N_r,M_2} \end{bmatrix} \tag{25}$$

where $\boldsymbol{X}'_j$ is the matrix of M_2 $\boldsymbol{X}_j$ matrices distributed on the main diagonal and the rest of the elements are zero. $\boldsymbol{R}''_{n_r,m}$ and $\boldsymbol{F}''_{n_r,m}$, respectively, can be expanded and written as

$$\begin{aligned} \boldsymbol{R}''_{n_r,m} &= \Big[r'_{n_r,m}, r'_{n_r,m+M_2}, \ldots, r'_{n_r,m+(n_{R_{n_r}}-1)M_2}, \ldots, \\ &\qquad r'_{n_r,m+(N_{R_{n_r}}-1)M_2}\Big]^T \\ \boldsymbol{F}''_{n_r,m} &= \begin{bmatrix} \boldsymbol{F}_j\left(m + \sum_{n=0}^{n_r-1} N'_{R_{n_r}}, :\right) \\ \boldsymbol{F}_j\left(m + M_2 + \sum_{n=0}^{n_r-1} N'_{R_{n_r}}, :\right) \\ \vdots \\ \boldsymbol{F}_j\left(m + (n_{R_{n_r}} - 1)M_2 + \sum_{n=0}^{n_r-1} N'_{R_{n_r}}, :\right) \\ \vdots \\ \boldsymbol{F}_j\left(m + (N_{R_{n_r}} - 1)M_2 + \sum_{n=0}^{n_r-1} N'_{R_{n_r}}, :\right) \end{bmatrix} \end{aligned} \tag{26}$$

$$\begin{aligned} \boldsymbol{Q}_j &: \boldsymbol{F}_j \boldsymbol{X}_j = \boldsymbol{R}_j \\ \boldsymbol{R}_j &= \left[\boldsymbol{R}'_1, \boldsymbol{R}'_2, \ldots, \boldsymbol{R}'_{n_r}, \ldots, \boldsymbol{R}'_{N_r}\right]^T \\ \boldsymbol{X}_j &= \left[\boldsymbol{X}'_{1,1}, \boldsymbol{X}'_{1,2}, \ldots, \boldsymbol{X}'_{1,m}, \ldots, \boldsymbol{X}'_{1,M_2}, \boldsymbol{X}'_{2,1}, \ldots, \boldsymbol{X}'_{n_{\mathrm{sp}},m}, \cdots \boldsymbol{X}'_{N_{\mathrm{sp}},M_2}\right]^T \\ \boldsymbol{F}_j &= \begin{bmatrix} \boldsymbol{F}'_{1,(1,1)} & \boldsymbol{F}'_{1,(1,2)} & \cdots & \boldsymbol{F}'_{1,(1,m)} & \cdots & \boldsymbol{F}'_{1,(1,M_2)} & \boldsymbol{F}'_{1,(2,1)} & \cdots & \boldsymbol{F}'_{1,(n_{\mathrm{sp}},m)} & \cdots & \boldsymbol{F}'_{1,(N_{\mathrm{sp}},M_2)} \\ \boldsymbol{F}'_{2,(1,1)} & \boldsymbol{F}'_{2,(1,2)} & \cdots & \boldsymbol{F}'_{2,(1,m)} & \cdots & \boldsymbol{F}'_{2,(1,M_2)} & \boldsymbol{F}'_{2,(2,1)} & \cdots & \boldsymbol{F}'_{2,(n_{\mathrm{sp}},m)} & \cdots & \boldsymbol{F}'_{2,(N_{\mathrm{sp}},M_2)} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \boldsymbol{F}'_{n_r,(1,1)} & \boldsymbol{F}'_{n_r,(1,2)} & \cdots & \boldsymbol{F}'_{n_r,(1,m)} & \cdots & \boldsymbol{F}'_{n_r,(1,M_2)} & \boldsymbol{F}'_{n_r,(2,1)} & \cdots & \boldsymbol{F}'_{n_r,(n_{\mathrm{sp}},m)} & \cdots & \boldsymbol{F}'_{n_r,(N_{\mathrm{sp}},M_2)} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ \boldsymbol{F}'_{N_r,(1,1)} & \boldsymbol{F}'_{N_r,(1,2)} & \cdots & \boldsymbol{F}'_{N_r,(1,m)} & \cdots & \boldsymbol{F}'_{N_r,(1,M_2)} & \boldsymbol{F}'_{N_r,(2,1)} & \cdots & \boldsymbol{F}'_{N_r,(n_{\mathrm{sp}},m)} & \cdots & \boldsymbol{F}'_{N_r,(N_{\mathrm{sp}},M_2)} \end{bmatrix} \\ \boldsymbol{F}'_{n_r,(n_{\mathrm{sp}},m)} &= \begin{bmatrix} f'_{1,1}(n_{\mathrm{sp}},m) & f'_{1,2}(n_{\mathrm{sp}},m) & \cdots & f'_{1,n'_E}(n_{\mathrm{sp}},m) & \cdots & f'_{1,N'_E}(n_{\mathrm{sp}},m) \\ f'_{2,1}(n_{\mathrm{sp}},m) & f'_{2,2}(n_{\mathrm{sp}},m) & \cdots & f'_{2,n'_E}(n_{\mathrm{sp}},m) & \cdots & f'_{2,N'_E}(n_{\mathrm{sp}},m) \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ f'_{n'_{R_{n_r}},1}(n_{\mathrm{sp}},m) & f'_{n'_{R_{n_r}},2}(n_{\mathrm{sp}},m) & \cdots & f'_{n'_{R_{n_r}},n'_E}(n_{\mathrm{sp}},m) & \cdots & f'_{n'_{R_{n_r}},N'_E}(n_{\mathrm{sp}},m) \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ f'_{N'_{R_{n_r}},1}(n_{\mathrm{sp}},m) & f'_{N'_{R_{n_r}},2}(n_{\mathrm{sp}},m) & \cdots & f'_{N'_{R_{n_r}},n'_E}(n_{\mathrm{sp}},m) & \cdots & f'_{N'_{R_{n_r}},N'_E}(n_{\mathrm{sp}},m) \end{bmatrix} \\ f'_{n'_{R_{n_r}},n'_E}(n_{\mathrm{sp}},m) &= \{0,1\} \end{aligned} \tag{20, 21}$$

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TABLE II
JSR UNDER DIFFERENT RELATIVE ESTIMATION ERRORS OF SAMPLING WIDTH

Estimation Error	-10%	-8%	-6%	-4%	-2%	0%	2%	4%	6%	8%	10%
JSR (dB)	-7.8	-9.4	-11.5	-14.7	-19.7	-25.0	-25.0	-25.0	-25.0	-25.0	-25.0

TABLE III
JSR UNDER DIFFERENT RELATIVE ESTIMATION ERRORS OF SAMPLING PERIOD

Estimation Error	-10%	-8%	-6%	-4%	-2%	0%	2%	4%	6%	8%	10%
JSR (dB)	3.8	1.8	-0.8	-3.5	-9.2	-25.0	-11.1	-6.1	-4.6	-3.7	-2.5

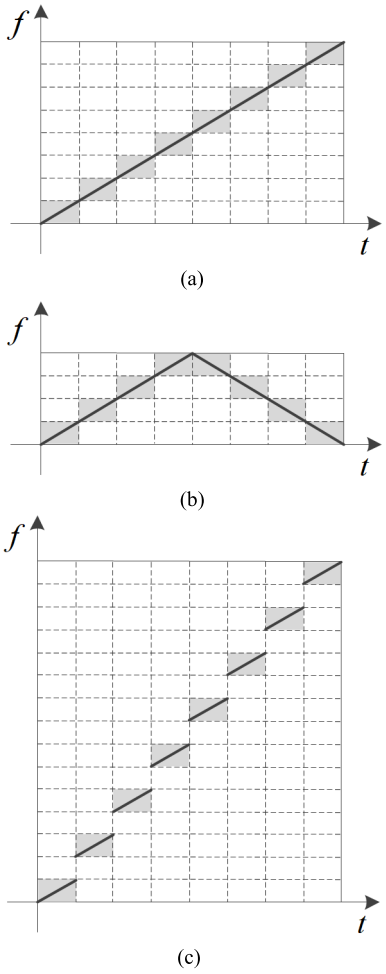


Fig. 7. Time–frequency diagram of the waveform. (a) LFM signal. (b) Triangular wave signal. (c) Frequency-stepped LFM signal.

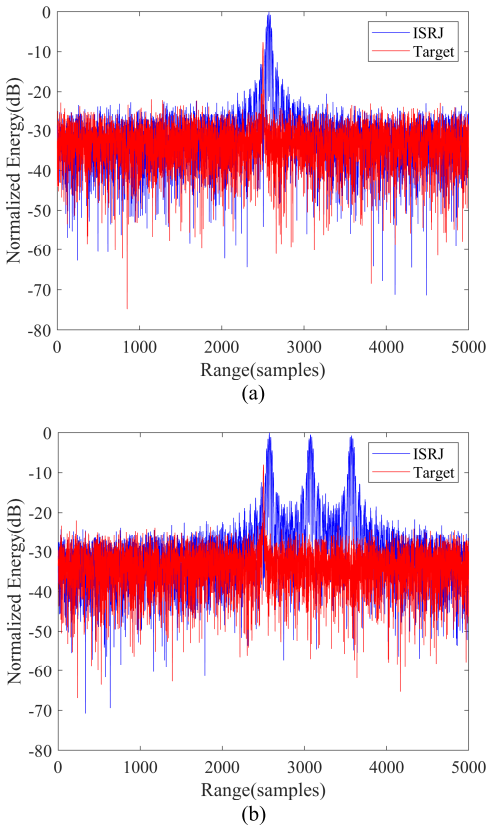


Fig. 8. 1-D range image of conventional SAR. (a) Image with D-ISRJ. (b) Image with R-ISRJ.

where $\boldsymbol{F}_j(m+(n_{R_{n_r}}-1)M_2+\sum_{n=0}^{n_r-1}N'_{R_{n_r}},:)$ represents the $(m+(n_{R_{n_r}}-1)M_2+\sum_{n=0}^{n_r-1}N'_{R_{n_r}})$th row vector in $\boldsymbol{F}_j$ shown in (20). Also, define $N'_{R_0}=0$.

Combining (19) with (26), it can be seen that the vector $[\,\boldsymbol{R}''_{n_r,1}\ \ \boldsymbol{R}''_{n_r,2}\ \ \cdots\ \ \boldsymbol{R}''_{n_r,m}\ \ \cdots\ \ \boldsymbol{R}''_{n_r,M_2}\,]$ in the deformed linear equation set $\boldsymbol{Q}'_j$ represents the echo in the n_rth reception window of the coding signal, which is the vector $\boldsymbol{R}_{n_r}$ in the original linear equation set $\boldsymbol{Q}_0$. Thus, when there is an error in the estimated parameters, the ISRJ signal $\boldsymbol{X}''_j$ after processing by the time-domain filter matrix $\boldsymbol{F}_t$ in (15) can be expressed as

$$\boldsymbol{X}''_j=\boldsymbol{F}_t\boldsymbol{R}'_j=\boldsymbol{F}_t\boldsymbol{F}'_j\boldsymbol{X}'_j \tag{27}$$

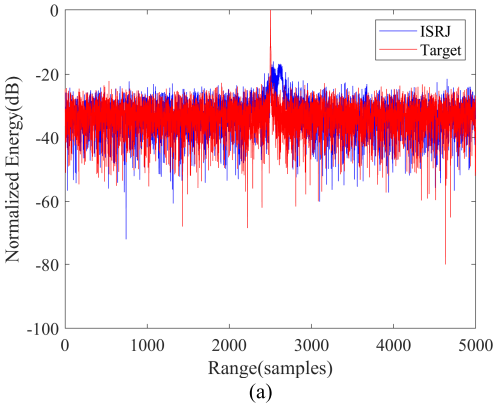
where the jamming parameters, jammer location, and coding signal form are known, and $\boldsymbol{F}_t$, $\boldsymbol{F}'_j$, and $\boldsymbol{X}'_j$ in (27) can be determined.

If the digital sampling frequency of the radar is f_s, the number of points L of a single subpulse signal after sampling can be calculated as

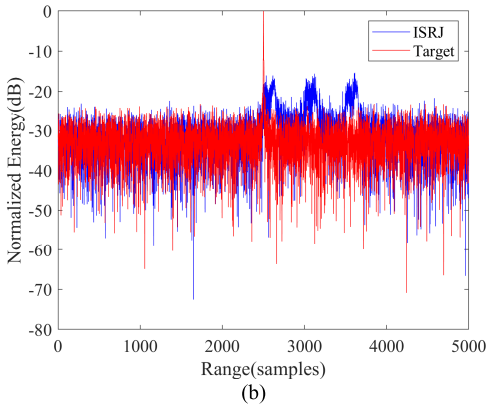
$$L=T_{\mathrm{sp}}\cdot f_s. \tag{28}$$

Then, after the segmented pulse compression processing, the strength of the ISRJ signal located at the position of the

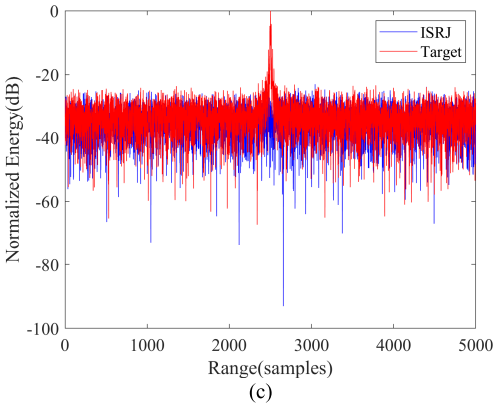
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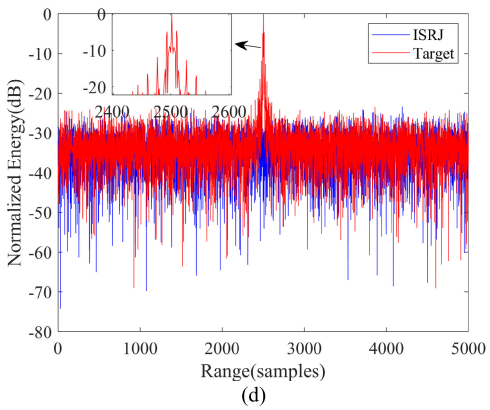
(a)



(b)

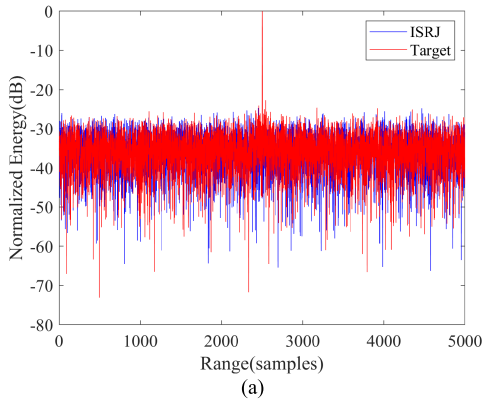


(c)

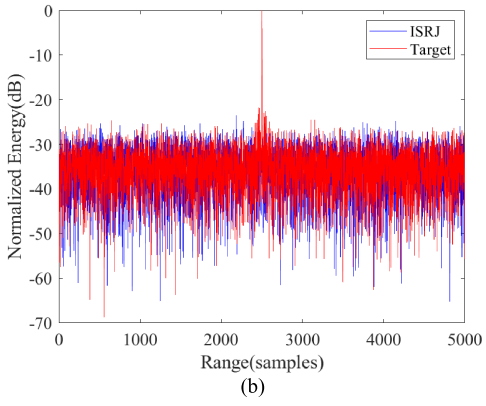


(d)

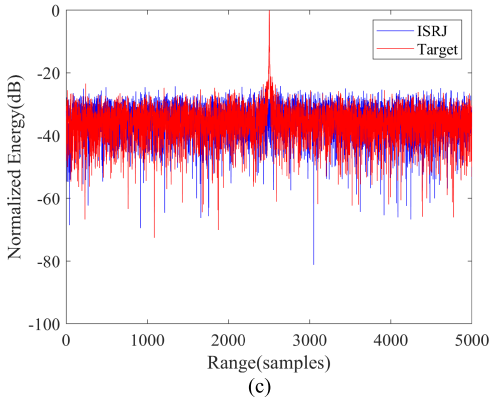
Fig. 9. 1-D range image of frequency-domain filtering SAR. (a) Image using LFM with D-ISRJ. (b) Image using LFM with R-ISRJ. (c) Image using frequency-stepped LFM with D-ISRJ. (d) Image using frequency-stepped LFM with R-ISRJ.



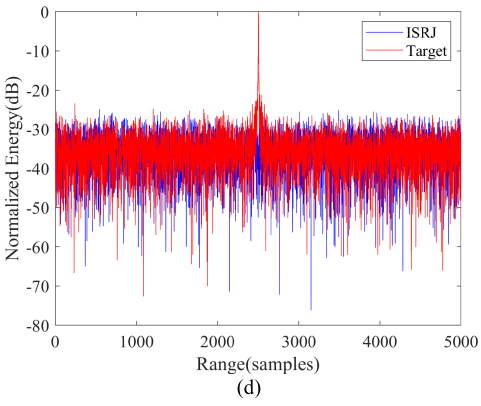
(a)



(b)



(c)



(d)

Fig. 10. 1-D range image of time-domain filtering SAR. (a) Image using LFM with D-ISRJ. (b) Image using LFM with R-ISRJ. (c) Image using triangular wave with D-ISRJ. (d) Image using triangular wave with R-ISRJ.

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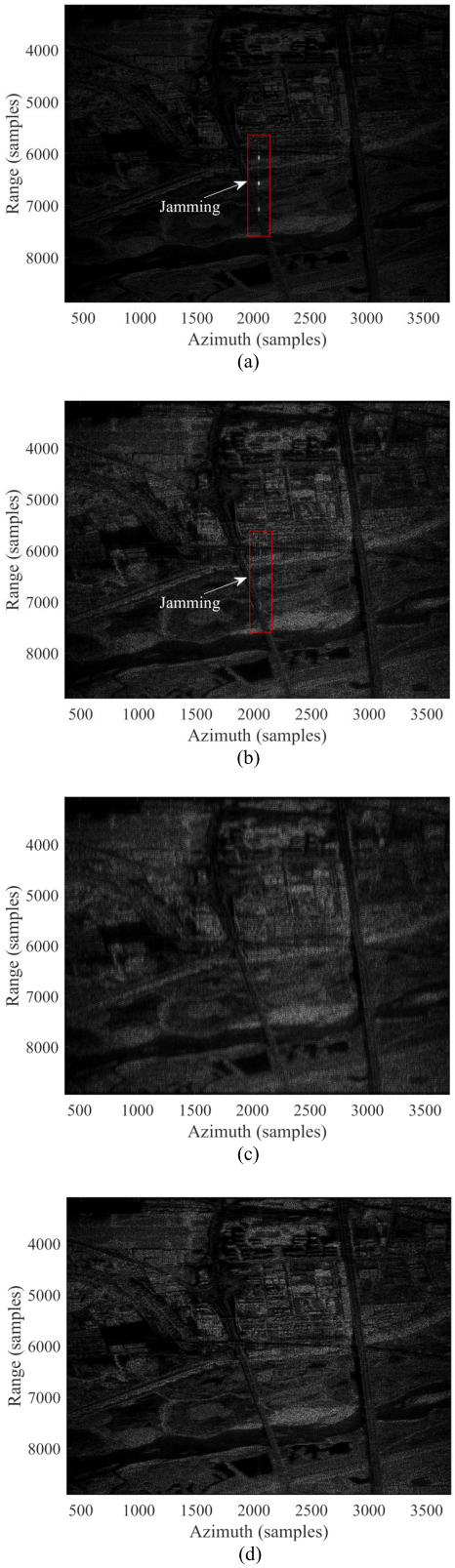


Fig. 11. Imaging results of area target. (a) Conventional SAR. (b) Frequency–domain filtering SAR using LFM. (c) Frequency-domain filtering SAR using frequency-stepped LFM. (d) Time-domain filtering SAR using LFM.

jammer can be expressed as

$$P_j(t_j) = \left| \sum_{n_{\rm sp}=1}^{N_{\rm sp}} \boldsymbol{H}_{n_{\rm sp}}^H \boldsymbol{J}_{n_{\rm sp}}(t_j) \right| \tag{29}$$

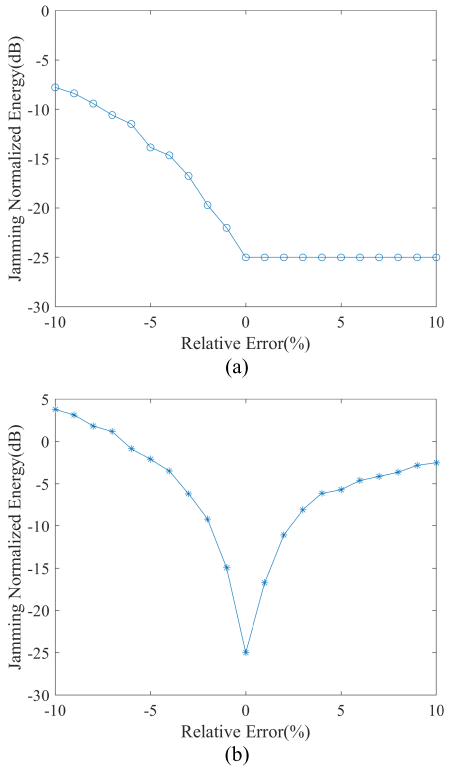


Fig. 12. Statistical curve of JSR. (a) Variation of JSR with estimation error of sampling width. (b) Variation of JSR with estimation error of sampling period.

where $\boldsymbol{H}_{n_{\rm sp}}$ is the matching filter function corresponding to subpulse $s_{n_{\rm sp}}$ with length L, $\boldsymbol{J}_{n_{\rm sp}}(t_j)$ is the jamming sampling signal with the center at the jammer position t_j and the length of L in the echo of subpulse $s_{n_{\rm sp}}$, and $\boldsymbol{J}_{n_{\rm sp}}(t_j)$ can be uniquely determined by $\boldsymbol{X}_j''$ in (27). In addition, the echo processing of removing a certain subpulse $s_{n_{\rm sp}}$ is equivalent to zeroing the matching filter function $\boldsymbol{H}_{n_{\rm sp}}$ of a certain subpulse $s_{n_{\rm sp}}$. Therefore, when the jamming parameters, jammer position, and coding signal form are known, the ISRJ residuals after processing by time-domain filtering anti-ISRJ method can be calculated by (29).

V. Simulations

In this section, imaging simulations of the time-domain filtering method and the frequency dimension filtering method are first performed to demonstrate the effectiveness of the anti-ISRJ method proposed in this article. Also, the simulations show the advantages of the time-domain filtering method over the frequency dimension filtering method. Then, a statistical analysis of the sensitivity of the proposed method to the jamming parameters will be performed. Finally, the point target imaging quality of coding SAR is compared with that of conventional SAR to show that the proposed method does not affect the focusing effect of the radar. And the parameters of all the simulations can be found in Table I.

In addition, Fig. 7 shows the three waveforms involved in all simulations, where Fig. 7(a) shows the conventional LFM waveform, Fig. 7(b) shows the triangular wave signal, and Fig. 7(c) shows the frequency-stepped LFM signal; the amount of frequency stepped by the subpulse is equal to the bandwidth of the subpulse.

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A. Anti-Jamming Simulation Results

Figs. 8–10 show the 1-D range images of conventional SAR, frequency-domain filtering SAR, and time-domain filtering SAR after adding D-ISRJ and R-ISRJ, respectively. Among them, Fig. 9(a) and (b) uses the frequency dimension filtering method based on the conventional LFM signal proposed in [31]. It can be seen that due to the partial transition band overlap in the spectrum between the subpulses in the LFM signal, the frequency-domain filtering method causes the existence of ISRJ residual. To solve the problem of jamming residuals, another frequency-domain filtering method based on frequency-stepped LFM signals was proposed in [32]. Since the spectrum of subpulses has no overlapping parts, the jamming echoes can be completely removed by frequency-domain filtering. However, the discontinuous frequency spectrum of the frequency-stepped LFM signal itself will cause the signal grating lobe problem shown in Fig. 9(c) and (d).

As shown in Fig. 10(a) and (b), since the time-domain filtering method obtains the echoes of each subpulse and performs the rejection of interfering signals in the time domain, the overlapping problem of the transition band spectrum of each subpulse in the LFM signal does not cause the problem of jamming residuals. Also, when encoding the intrapulse signal, the jamming parameter as a priori information is used to assign the edge band to the interfered subpulse in advance. In this way, the frequency band occupied by the undisturbed subpulse is still continuous, which will not cause the grating lobe problem but will reduce the resolution of SAR in the range dimension. In addition, the anti-ISRJ results of the triangular wave signal shown in Fig. 10(c) and (b) demonstrate the advantage of high waveform design freedom of the proposed method in this article.

The area target imaging results of conventional SAR, frequency-domain filtering SAR, and time-domain filtering SAR are shown in Fig. 11. It can be seen that the ISRJ in the time-domain filtering SAR is effectively suppressed. The frequency-domain filtering SAR based on LFM signal has jamming residual, while the frequency-domain filtering SAR image based on a frequency-stepped LFM signal becomes blurred due to the grating lobe problem.

B. Sensitivity Analysis of the Proposed Method to Jamming Parameters

To evaluate the influence of the two key parameters of jamming sampling width and sampling period on the anti-ISRJ performance, a group of parameter sensitivity analysis experiments is conducted in this section. As shown in Figs. 12, 13 and Tables II, III, the time-domain filtering method is more sensitive to the jamming sampling period than to the jamming sampling width. When the relative estimation error of the sampling period is $\pm 5\%$, the jamming signal is quite obvious. It can be seen from Fig. 12(a) and Table II that when the relative estimation error of the sampling width is positive, the availability of the anti-ISRJ method introduced in this article is not affected. This is because when the actual sampling width of the jammer is less than the subpulsewidth, the rejected complete subpulse already contains the partially sampled signal.

C. Analysis of the Imaging Quality of the Proposed Method

Since the time-domain filtering method is to decode the complete echo of each subpulse from the mixed echo of all pulses within a pulse coding repetition interval and perform the imaging process. Consequently, it is essential to judge whether

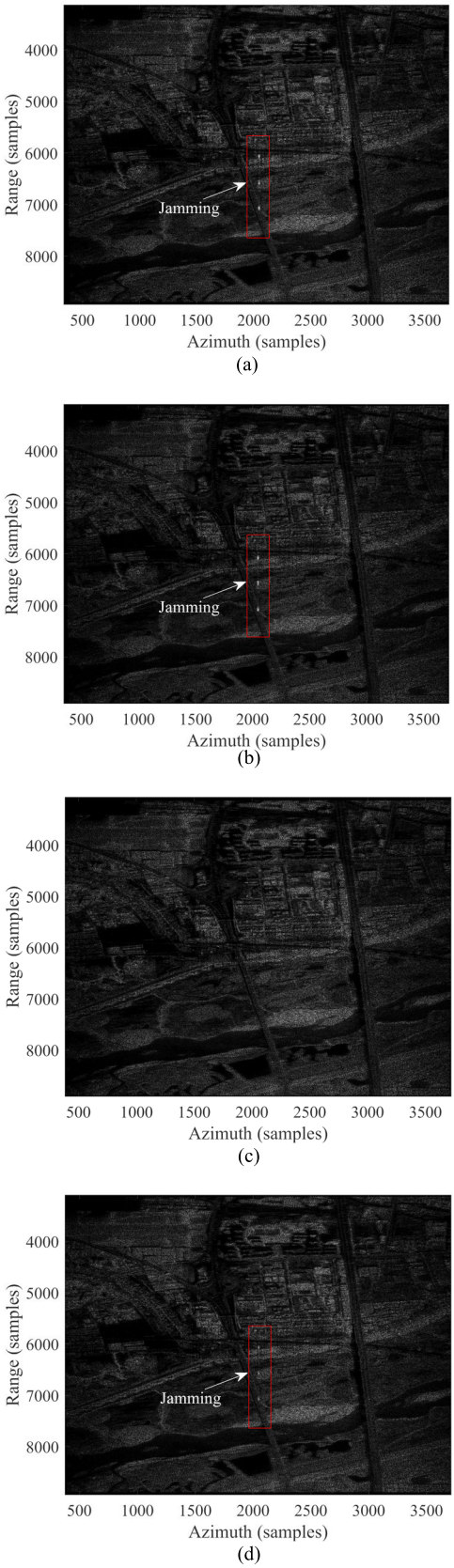


Fig. 13. Imaging results of time-domain filtering SAR under different estimation errors. (a) Sampling period with a relative error of 5%. (b) Sampling period with a relative error of −5%. (c) Sampling width with a relative error of 5%. (d) Sampling width with a relative error of −5%.

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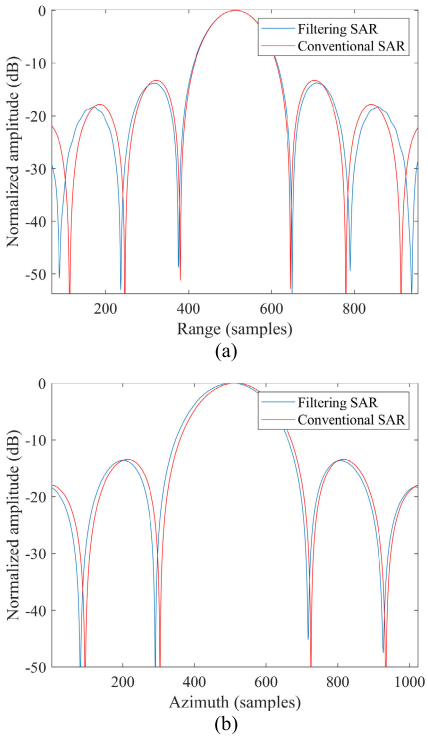


Fig. 14. Point target imaging results. (a) Range profile. (b) Azimuth profile.

TABLE IV
IMAGING QUALITY OF CONVENTIONAL SAR

Symbol	Parameter	Value
ρ_a	resolution in azimuth	3.18 m
ρ_r	resolution in range	2.21 m
$PSLR_a$	azimuth PSLR	-13.41 dB
$PSLR_r$	range PSLR	-13.26 dB
$ISLR_a$	azimuth ISLR	-11.86 dB
$ISLR_r$	range ISLR	-11.50 dB

TABLE V
IMAGING QUALITY OF TIME-DOMAIN FILTERING SAR

Symbol	Parameter	Value
ρ_a	resolution in azimuth	3.26 m
ρ_r	resolution in range	2.25 m
$PSLR_a$	azimuth PSLR	-13.76 dB
$PSLR_r$	range PSLR	-13.78 dB
$ISLR_a$	azimuth ISLR	-12.14 dB
$ISLR_r$	range ISLR	-11.27 dB

the proposed method will reduce the radar imaging quality. Fig. 14 shows the range profiles and azimuth profiles of two kinds of SAR. Also, Tables IV and V record the indexes used to evaluate the two types of radar imaging results, respectively. It can be seen that the values of the 2-D resolution, 2-D integral sidelobe ratio, and 2-D peak sidelobe ratio of the time-domain filtering SAR are consistent with those of the conventional SAR. Therefore, the method proposed in this article does not reduce the focusing ability of the radar on the target.

VI. CONCLUSION

The article proposes an intrapulse joint interpulse coding waveform and a corresponding time-domain filtering method to counter ISRJ. Compared with the frequency-domain filtering method, the proposed algorithm has a better anti-ISRJ effect and no grating lobe problem. In addition, since the coding method does not require the waveform of the subpulse, the proposed method has a high degree of freedom in waveform design. Since the method proposed in this article discards the echo of the subpulse sampled by the jammer, the echo gain of the target signal after pulse compression is lost. In addition, the anti-jamming performance of the method proposed in this article depends on the accuracy of the obtained jamming parameters, so the accurate estimation of the jamming parameters is the focus of the next research.

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